Interpret inequalities

1 Write the following sentences using mathematical symbols:

a n is greater than 17.

b n is less than 10.

c w is less than $\frac{2}{3}$.

d y is greater than or equal to 12.

e The product of 6 and x is less than or equal to -42.

f The sum of 5 and the product of 4 and x is greater than or equal to 20.

2 Write the following inequalities in words:

a $p < 9$ **b** $f > 14$ **c** $m \leq 23$ **d** $a \geq 88$

Solve inequalities

3 State the largest integer value x can have if $x \leq -2$.

4 State the smallest integer value x can have if $x > 19$.

5 State whether the following are solutions of $k \leq 15$:

a $k = 14$ **b** $k = -15$ **c** $k = 15$ **d** $k = 16$

6 Consider the inequality: $a + 9 < 12$.

a Find the value of the left-hand side of the inequality when $a = 1$.

b Is $a = 1$ a solution of $a + 9 < 12$?

7 Consider the inequality: $12 - y > 10$.

a Find the value of the left-hand side of the inequality when $y = 0$.

b Is $y = 0$ a solution of $12 - y > 10$?

8 Consider the inequality: $9b \geq 18$.

a Find the value of the left-hand side of the inequality when $b = 3$.

b Is $b = 3$ a solution of $9b \geq 18$?

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9 State whether the following are solutions of $x + 2 > 6$:

a $x = 4$ **b** $x = 5$ **c** $x = 3$ **d** $x = -9$

10 Consider the inequality $-2x > 10$. Will the direction of the inequality symbol be reversed when solving this inequality? Explain your answer.

11 Sally attempted to solve the inequality $-18 - 6x > 30$, but has made a mistake. Her working is shown below:

$$\begin{aligned} \text{Step 1}: &\quad -18 - 6x > 30 \\ \text{Step 2}: &\quad -6x > 48 \\ \text{Step 3}: &\quad x > -8 \end{aligned}$$

What was Sally's mistake?

12 Solve the following inequalities:

a $7y > 42$

b $8x \leq -48$

c $-9x < 63$

d $-4x \leq -16$

e $x - 1 < 15$

f $7 - x < 2$

g $2 \leq x - 6$

h $8x + 16 > 64$

i $7x - 49 \geq 42$

j $7x + 5 > x + 35$

k $-3(x + 9) \geq -6$

l $-9(x + 6) \geq -27$

m $23 - 3k \leq 2k + 3$

n $-4 - 4x > -7x + 2$

o $\dfrac{x}{3} \geq 7$

p $\dfrac{x}{-4} \geq 8$

q $\dfrac{x + 9}{8} \geq 9$

r $2 - \dfrac{1}{2}x > 5$

s $\dfrac{x}{2} + \dfrac{x}{3} \geq 5$

t $\dfrac{x}{9} - \dfrac{x}{7} < 0$

u $\dfrac{a}{6} < 11$

v $\dfrac{a}{5} + 9 > 9$

w $91 - 4k \leq 5k + 10$

13 Write the following sentences using mathematical symbols and solve:

a The sum of 3 groups of p, and 9, is less than 24.

b The sum of 5 groups of x, and 3 is at least 23.

c Six more than the value of x is at least seven.

d Half of x is no more than five.

e Negative four groups of x is less than three.

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15 Consider the following: "5 groups of p minus 10 is no more than 10".

a Write the sentence using mathematical symbols and solve for p.

b Find the largest largest integer value of p that satisfies this condition.

16 Consider the inequality: $6\left(x+6\right) \leq 42$.

a Solve the inequality.

b State whether the following are solutions of: $6\left(x+6\right) \leq 42$.

i $x=-1$ **ii** $x=2$ **iii** $x=0$ **iv** $x=1$

17 Consider the inequality: $30+3x>30$.

a Solve the inequality.

b State whether the following are solutions of: $30+3x>30$.

i $x=1$ **ii** $x=0$ **iii** $x=3$ **iv** $x=6$

18 Consider the inequality: $3\left(3x-4\right) \geq 15$.

a Solve the inequality.

b State whether the following are solutions of: $3\left(3x-4\right) \geq 15$.

i $x=0$ **ii** $x=3$ **iii** $x=5$ **iv** $x=-3$

Applications

19 Write the following statement as an inequality in terms of T: The temperature T inside a freezer is always below $-15^{\circ}\mathrm{C}$.

20 Derek is saving up to buy a tablet that is selling for $\$890$. He has $\$760$ in his bank account and expects a sum of money for his birthday next month. If the amount he is to receive for his birthday is represented by x, write an inequality that models the situation where he is able to afford the tablet.

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21 The table shows the time, t, of six swimmers in a 1500 m race. Only those who achieve a time of below 23 minutes qualify for the final. Find all the swimmers who qualify for the final.

Swimmer	Time, t (minutes)
Adam	32
Vincent	22
Yuri	28
Aaron	30
Neville	20
David	26

22 The table shows the distances, d, achieved by six javelin throwers. Only those who achieve a distance of more than 75 m qualify for the final. Find all the throwers who qualify for the final.

Thrower	Distance, d (metres)
Tom	63
Quentin	67
Bart	74
Bob	85
Tobias	90
Adam	71

23 To get a grade of C, Luke must obtain a total score of at least 300 over his four exams. So far he has taken the first three exams and achieved scores of 65, 51, and 97. If x represents what he must score on the last exam to get a C or better, write an inequality and solve for x.

24 Sandy has a budget for school stationery of $\$46$, but has already spent $\$18.10$ on books and folders. Let p represent the amount that Sandy can spend on other stationery. Write an inequality that shows how much she can spend on other stationery, and solve for p.

25 Ursula was given $\$60$ for a birthday present. This present, along with earnings from a Saturday job, are being set aside for a mountain bike. The job pays $\$6.50$ per hour, and the bike costs $\$361$.

a Write and solve an inequality to find the minimum number of hours that Ursula needs to work to be able to buy the mountain bike, where h represents the number of hours worked.

b If Ursula can only work her job for a whole number of hours, find the minimum number of hours she must work to afford her bike.

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26 Ivan wants to save up enough money so that he can buy a new sports equipment set, which costs $\$60.00$. Ivan has $\$21.40$ that he has saved from his birthday. In order to make more money, he plans to wash neighbors' windows for $\$4$ per window.

a Let w be the number of windows that Ivan washes. Write and solve an inequality to find the minimum number of windows he needs to wash in order to afford the equipment.

b Find the minimum whole number of windows that Ivan must wash to be able to afford the equipment.

27 Charlie is saving up to buy a laptop that is selling for $\$740$. He has $\$520$ in his bank account and expects a nice sum of money for his birthday next week.

a If the amount he is to receive for his birthday is represented by x, write an inequality that models the situation where he is able to afford the laptop.

b Would he have enough to buy the laptop if his parents were to give him $\$180$ for his birthday?

28 When breeding certain types of fish, it is recommended that the number of female fish is more than triple the number of male fish.

a If the number of females is represented by f and the number of males is represented by m, write an inequality that represents the recommended relationship between f and m.

b Find a number of female and male fish that would satisfy the recommendation.

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